



# **AWS Academy Lab Project – Cloud Web Application Builder**

By

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# Business scenario overview

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- Provide problem or opportunity statement.

The scenario involves planning, designing, building and deploying a web application to the AWS Cloud in a manner consistent with the AWS Well-Architected Framework's best practices. During the peak admissions period, the application must support thousands of users and offer high availability, scalability, load balancing, security and performance.

# Business scenario overview

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- Describe solution requirements.
  1. Create an architectural diagram depicting the various AWS services and how they interact with each other.
  2. Use the AWS Pricing Calculator to estimate the cost of using the services.
  3. Deploy a functional web application that runs on a single virtual machine and is supported by a relational database.
  4. Design a web application with separate layers, such as the web server and database.
  5. Configure a virtual network to host a publicly accessible and secure web application.
  6. Deploy a web application with load distributed across multiple web servers.
  7. Configure the appropriate network security settings for the web servers and database.
  8. Implement high availability and scalability in the deployed solution.



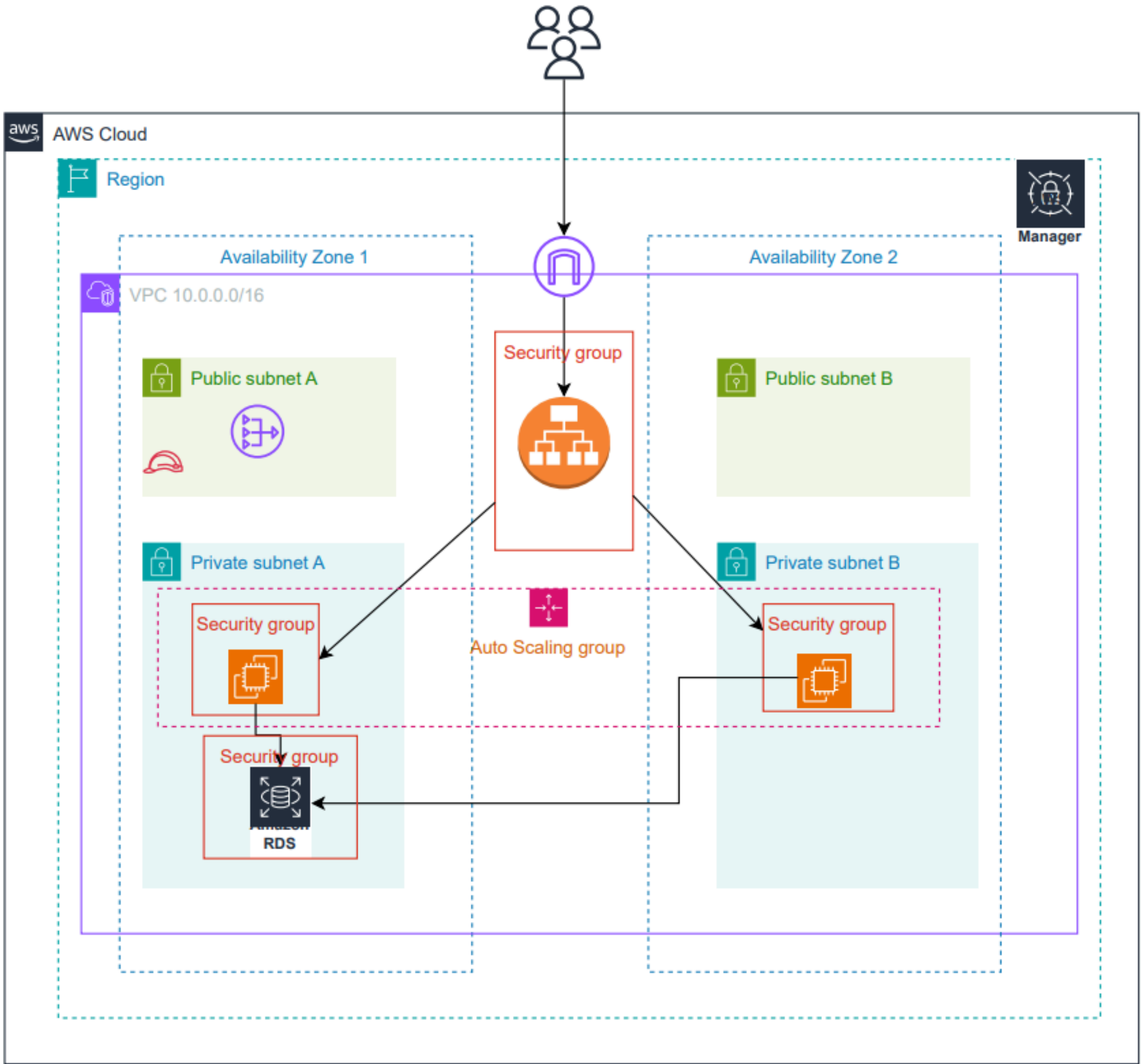
# Phase 1: Planning the design and estimating cost

## Task 1: Creating an architectural diagram

Plan the design and implementation by creating an architecture diagram for the desired solution.



# Architecture diagram of the solution



# Cost estimation

|                  |              |                       |
|------------------|--------------|-----------------------|
| Estimate summary |              |                       |
| Upfront cost     | Monthly cost | Total 12 months cost  |
| 0.00 USD         | 79.87 USD    | 958.44 USD            |
|                  |              | Includes upfront cost |

## Detailed Estimate

|                                    |                                                                                                                                                                                                                                                                                                                                        |                       |              |              |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------|--------------|
| Name                               | Group                                                                                                                                                                                                                                                                                                                                  | Region                | Upfront cost | Monthly cost |
| Amazon EC2                         | -                                                                                                                                                                                                                                                                                                                                      | US East (N. Virginia) | 0.00 USD     | 14.02 USD    |
| Status                             | -                                                                                                                                                                                                                                                                                                                                      |                       |              |              |
| Description:                       | -                                                                                                                                                                                                                                                                                                                                      |                       |              |              |
| Config summary                     | Tenancy (Shared Instances), Operating system (Ubuntu Pro), Workload (Consistent, Number of instances: 2), Advance EC2 instance (t3a.micro), Pricing strategy ( 1yr No Upfront), Enable monitoring (disabled), DT Inbound: Not selected (0 TB per month), DT Outbound: Not selected (0 TB per month), DT Intra-Region: (0 TB per month) |                       |              |              |
| Name                               | Group                                                                                                                                                                                                                                                                                                                                  | Region                | Upfront cost | Monthly cost |
| Amazon Virtual Private Cloud (VPC) | -                                                                                                                                                                                                                                                                                                                                      | US East (N. Virginia) | 0.00 USD     | 7.30 USD     |
| Status                             | -                                                                                                                                                                                                                                                                                                                                      |                       |              |              |
| Description:                       | -                                                                                                                                                                                                                                                                                                                                      |                       |              |              |
| Config summary                     | Number of In-use public IPv4 addresses (2)                                                                                                                                                                                                                                                                                             |                       |              |              |
| Name                               | Group                                                                                                                                                                                                                                                                                                                                  | Region                | Upfront cost | Monthly cost |
| Elastic Load Balancing             | -                                                                                                                                                                                                                                                                                                                                      | US East (N. Virginia) | 0.00 USD     | 22.27 USD    |
| Status                             | -                                                                                                                                                                                                                                                                                                                                      |                       |              |              |
| Description:                       | -                                                                                                                                                                                                                                                                                                                                      |                       |              |              |
| Config summary                     | Number of Application Load Balancers (1)                                                                                                                                                                                                                                                                                               |                       |              |              |
| Name                               | Group                                                                                                                                                                                                                                                                                                                                  | Region                | Upfront cost | Monthly cost |
| Amazon RDS for MySQL               | -                                                                                                                                                                                                                                                                                                                                      | US East (N. Virginia) | 0.00 USD     | 35.88 USD    |
| Status                             | -                                                                                                                                                                                                                                                                                                                                      |                       |              |              |
| Description:                       | -                                                                                                                                                                                                                                                                                                                                      |                       |              |              |
| Config summary                     | Storage amount (20 GB), Storage for each RDS instance (General Purpose SSD (gp2)), Nodes (1), Instance type (db.t4g.micro), Utilization (On-Demand only) (100 %Utilized/Month), Deployment option (Single-AZ), Pricing strategy (OnDemand)                                                                                             |                       |              |              |
| Name                               | Group                                                                                                                                                                                                                                                                                                                                  | Region                | Upfront cost | Monthly cost |
| AWS Secrets Manager                | -                                                                                                                                                                                                                                                                                                                                      | US East (N. Virginia) | 0.00 USD     | 0.40 USD     |
| Status                             | -                                                                                                                                                                                                                                                                                                                                      |                       |              |              |
| Description:                       | -                                                                                                                                                                                                                                                                                                                                      |                       |              |              |

# Phase 2: Creating a basic functional web application

## Task 1: Creating a virtual network

Create networking resources such as a virtual private cloud (VPC) and subnets.

**Create VPC** [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as subnets, Elastic IP addresses, and Amazon EC2 instances.

**VPC settings**

**Resources to create** [Info](#)  
Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

**Name tag auto-generation** [Info](#)  
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate  
xyz-web-vpc

**IPv4 CIDR block** [Info](#)  
Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16 65,536 IPs  
CIDR block size must be between /16 and /28.

**IPv6 CIDR block** [Info](#)

☒ No IPv6 CIDR block  
☐ Amazon-provided IPv6 CIDR block

**Tenancy** [Info](#)  
Default

[VPC](#) > [Your VPCs](#) > Create VPC

The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

0 | **2**

**Number of private subnets** [Info](#)  
The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0 | **2** | 4

► **Customize subnets CIDR blocks**

**NAT gateways (\$)** [Info](#)  
Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway.

None | In 1 AZ | **1 per AZ**

**VPC endpoints** [Info](#)  
Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

None | **S3 Gateway**

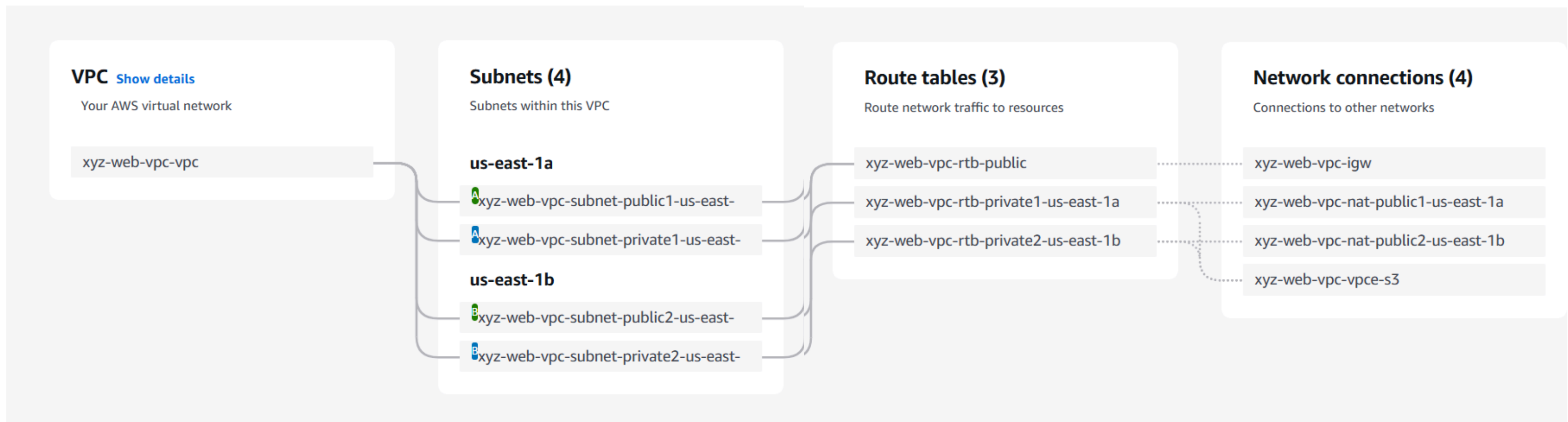
**DNS options** [Info](#)

☒ Enable DNS hostnames  
☒ Enable DNS resolution

# Phase 2: Creating a basic functional web application

## Task A: Create VPC Components

Now we will create subnets, an internet gateway, and route tables for our VPC. No need to change the VPC configuration



# Phase 2: Creating a basic functional web application

## Task 2: Creating a virtual machine

EC2 > Instances > Launch an instance

**Name and tags** [Info](#)

Name

xyz-ec2 [Add additional tags](#)

▼ **Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux macOS **Ubuntu** Windows Red Hat SUSE Linux Debian

aws Mac ubuntu® Microsoft Red Hat SUSE debian

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

**Amazon Machine Image (AMI)**

▼ **Summary**

**Number of instances** [Info](#)

1

**Software Image (AMI)**

Canonical, Ubuntu, 24.04, amd64...[read more](#)

ami-0360c520857e3138f

**Virtual server type (instance type)**

t3.micro

**Firewall (security group)**

New security group

**Storage (volumes)**

1 volume(s) - 8 GiB

**Free tier:** In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where

Create a  
Ubuntu  
EC2



created a new key pair and downloaded it

aws

Search

[Alt+S]

EC2

Instances

Launch an instance

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand Linux base pricing: 0.0104 USD per Hour

On-Demand RHEL base pricing: 0.0392 USD per Hour On-Demand Windows base pricing: 0.0196 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

xyz-web

Create new key pair

Network settings

Info

Edit

Network

Info

vpc-07f5c2de536a8232d

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Downloads

xyz-web.ppk

Open file

Building microservices and CI\_CD Pipeline.drawio

Open file

See more

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-0360c520857e3138f

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where

Cancel

Launch instance

Preview code

Activate Windows

Go to Settings to activate Windows.



# Configuring the development environment

1. Go to AWS Cloud9 Services
2. Choose Create Environment
3. Enter the Name and Description
4. Select New EC2 instance for Cloud9 Environment
5. Instance Type is t3.micro
6. For the platform, choose Linux Server
7. In Network Settings:  
Choose SSH connection

## Create environment [Info](#)

### Details

#### Name

Limit of 60 characters, alphanumeric, and unique per user.

#### Description - optional

Limit 200 characters.

#### Environment type [Info](#)

Determines what the Cloud9 IDE will run on.



##### New EC2 instance

Cloud9 creates an EC2 instance in your account. The configuration of your EC2 instance cannot be changed by Cloud9 after creation.



##### Existing compute

You have an existing instance or server that you'd like to use.

## Network settings [Info](#)

### Connection

How your environment is accessed.



#### AWS Systems Manager (SSM)

Accesses environment via SSM without opening inbound ports (no ingress).



#### Secure Shell (SSH)

Accesses environment directly via SSH, opens inbound ports.

### ▼ VPC settings [Info](#)

#### Amazon Virtual Private Cloud (VPC)

The VPC that your environment will access. To allow the AWS Cloud9 environment to connect to its EC2 instance, attach an internet gateway (IGW) to your VPC. [Create new VPC](#)

Name - xyz-web-vpc-vpc



#### Platform [Info](#)

This will be installed on your EC2 instance. We recommend Amazon Linux 2023.



#### Timeout

How long Cloud9 can be inactive (no user input) before auto-hibernating. This helps prevent unnecessary charges.



# Create Security Group rules

Additional charges apply when outside of free tier allowance

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

Test-EC2-SG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and .\_-:/()#,@[]+=&;{}!\$\*

Description - required

Info

TEST

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 80, 0.0.0.0/0)

Remove

Type

Info

HTTP

▼

Protocol

Info

TCP

Port range

Info

80

Source type

Info

Anywhere

▼

Source

Info

Q Add CIDR, prefix list or security group

Description - optional

Info

e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 3306, sg-0e17c6e9921efeb79)

Remove

Type

Info

MYSQL/Aurora

▼

Protocol

Info

TCP

Port range

Info

3306

Source type

Info

Custom

▼

Source

Info

Q Add CIDR, prefix list or security group

sg-0e17c6e9921efeb79 X

Description - optional

Info

e.g. SSH for admin desktop



### IAM instance profile | [Info](#)

LabInstanceProfile

arn:aws:iam::302531438417:instance-profile/LabInstanceProfile

### User data - *optional* | [Info](#)

Upload a file with your user data or enter it in the field.

 **Choose file**

```
#!/bin/bash -xe
apt update -y
apt install nodejs unzip wget npm mysql-server -y
#wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-200-ACCAP1-1-DEV/code.zip -P /home/ubuntu
wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-200-ACCAP1-1-91571/1-lab-capstone-project-1/code.zip -P /home/ubuntu
cd /home/ubuntu
unzip code.zip -x "resources/codebase_partner/node_modules/*"
cd resources/codebase_partner
npm install aws aws-sdk
mysql -u root -e "CREATE USER 'nodeapp' IDENTIFIED WITH mysql_native_password BY 'student12'";
mysql -u root -e "GRANT all privileges on *.* to 'nodeapp'@'%';"
mysql -u root -e "CREATE DATABASE STUDENTS;"
```

☐ User data has already been base64 encoded

Add IAM profile and copy the  
Script UserdataScript-phase-2.sh  
And paste it.

# Phase 2: Creating a basic functional web application

- Task 3: Testing the deployment

Add a few dummy entries



# XYZ University

[Home](#)  
[Students list](#)

## All students

| Name  | Address | City  | State | Email          | Phone      |                      |
|-------|---------|-------|-------|----------------|------------|----------------------|
| Rawan | Amman   | Amman | amman | test@gmail.com | 0766252742 | <a href="#">edit</a> |

Add a new student

- Details
- Status and alarms
- Monitoring
- Security
- Networking
- Storage
- Tags

▼ Instance summary [Info](#)

|                                                                                                                            |                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Instance ID</b><br> i-03140ab55843aed98 | <b>Public IPv4 address</b><br> 44.211.94.88   <a href="#">open address</a>  | <b>Private IPv4 addresses</b><br> 10.0.8.247                                                                                                                                         |
| <b>IPv6 address</b><br>—                                                                                                   | <b>Instance state</b><br> <b>Running</b>                                                                                                                       | <b>Public DNS</b><br><br>ec2-44-211-94-88.compute-1.amazonaws.com   <a href="#">open address</a>  |



# Welcome

Use this app to keep track of your student inquiries

[List of students](#)

# Phase 3: Decoupling the application components

- **Task 1: Changing the VPC configuration**

No need to change

- **Task 2: Creating and configuring the Amazon RDS database**

Create an Amazon Relational Database Service (Amazon RDS) database that runs a MySQL engine. You can choose to create a provisioned instance or run it serverless.

Create database [Info](#)

Choose a database creation method

☒ Standard create  
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create  
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible)  


☐ Aurora (PostgreSQL Compatible)  


☒ MySQL  


Templates

Choose a sample template to meet your use case.

☐ Production  
Use defaults for high availability and fast, consistent performance.

☐ Dev/Test  
This instance is intended for development use outside of a production environment.

☒ Free tier  
Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. [Info](#)

Availability and durability

Deployment options [Info](#)

Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the [Amazon RDS service level agreement \(SLA\)](#) [↗](#).

☐ Multi-AZ DB cluster deployment (3 instances)  
Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:

- 99.95% uptime
- Redundancy across Availability Zones
- Increased read capacity
- Reduced write latency

☐ Multi-AZ DB instance deployment (2 instances)  
Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:

- 99.95% uptime
- Redundancy across Availability Zones

☒ Single-AZ DB instance deployment (1 instance)  
Creates a single DB instance without standby instances. This setup provides:

- 99.5% uptime
- No data redundancy



## In Credential Settings

Enter your  
Master Username  
Master Password

Aurora and RDS > Databases > Create database

**Master username** [Info](#)  
Type a login ID for the master user of your DB instance.

nodeapp

1 to 16 alphanumeric characters. The first character must be a letter.

**Credentials management**  
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - *most secure*  
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed  
Create your own password or have RDS create a password that you manage.

☐ Auto generate password  
Amazon RDS can generate a password for you, or you can specify your own password.

**Master password** [Info](#)  
.....

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' " @

**Confirm master password** [Info](#)  
.....

```
mysql -u root -e "CREATE USER 'nodeapp' IDENTIFIED WITH mysql_native_password BY 'student12';"  
mysql -u root -e "GRANT all privileges on *.* to 'nodeapp'@'%';"  
mysql -u root -e "CREATE DATABASE STUDENTS;"
```



**Create a subnet group**

**Create a security group**

Aurora and RDS > Databases > Create database

☒ Don't connect to an EC2 compute resource  
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ Connect to an EC2 compute resource  
Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) Info

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

xyz-web-vpc-vpc (vpc-02f2bbfad78d58798)  
4 Subnets, 2 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group Info

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

Create new DB Subnet Group

Public access Info

☐ Yes  
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ No  
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) Info

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☐ Choose existing  
Choose existing VPC security groups

☒ Create new  
Create new VPC security group

Activate Windows  
Go to Settings to activate Windows

▼ Additional configuration

Database options, encryption turned on, backup turned on, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

Database options

Initial database name Info

STUDENTS

If you do not specify a database name, Amazon RDS does not create a database.

DB parameter group Info

default.mysql8.0

Option group Info

default:mysql-8-0

Backup

☒ Enable automated backup  
Creates a point-in-time snapshot of your database

# Phase 3: Decoupling the application components

- **Task 3: Configuring the development environment** done in the previous step

- **Task 4: Provisioning Secrets Manager**

update to new

```
https://us-east-1.console.aws.amazon.com/cloud9/ide/547a080a7eb541cb92243de59de62c92?region=us-east-1

File Edit Find View Go Run Tools Window Support Preview Run

Go to Anything (Ctrl-P)
webenv - /home/ec2-user/
.c9
README.md

aws

voclabs:~/environment $ aws secretsmanager create-secret \
> --name Mydbsecret \
> --description "Database secret for web app" \
> --secret-string "{\"user\":\"nodeapp\",\"password\":\"student12\",\"host\":\"database-1.clvflfwpvhti.us-east-1.rds.amazonaws.com\",\"db\":\"STUDENTS\"}"

An error occurred (ResourceExistsException) when calling the CreateSecret operation: The operation failed because the secret Mydbsecret already exists.
voclabs:~/environment $ aws secretsmanager update-secret \
> --secret-id Mydbsecret \
> --secret-string "{\"user\":\"nodeapp\",\"password\":\"student12\",\"host\":\"database-1.clvflfwpvhti.us-east-1.rds.amazonaws.com\",\"db\":\"STUDENTS\"}"
{
  "ARN": "arn:aws:secretsmanager:us-east-1:302531438417:secret:Mydbsecret-WnU5WU",
  "Name": "Mydbsecret",
  "VersionId": "f6ddfd14-cc74-457b-8bf3-826068e23f50"
}
voclabs:~/environment $
```



# Cloud9

Script-1:

# Use following script to create secret in secret manager for accessing your database via the web application  
# Replace the values for below placeholders with the actual values you have used to configure the service

#<RDS Endpoint>

#<password>

#<username>

#<dbname>

aws secretsmanager create-secret \

--name Mydbsecret \

--description "Database secret for web app" \

--secret-string "{\"user\":\"<username>\",\"password\":\"<password>\",\"host\":\"<RDS Endpoint>\",\"db\":\"<dbname>\"}"

# Phase 3: Decoupling the application components

- **Task 5: Provisioning a new instance for the web server**  
Create a new virtual machine to host the web application.

Instances (2/9) [Info](#)

Find Instance by attribute or tag (case-sensitive)

Instance state = running X Clear filters

Connect

Instance state ▼

Actions ▼

Launch instances ▼

All states ▼

|                                     | Name                                            | Instance ID         | Instance state ▼ | Instance type ▼ | Status check | Alarm status                  |
|-------------------------------------|-------------------------------------------------|---------------------|------------------|-----------------|--------------|-------------------------------|
| <input type="checkbox"/>            | aws-cloud9-webenv-547a080a7eb541cb92243de59d... | i-04bb49cd881aff132 | Running          | t3.micro        | Initializing | <a href="#">View alarms +</a> |
| <input checked="" type="checkbox"/> | Host ec2                                        | i-0c02ae43c34ff9884 | Running          | t3.micro        | Initializing | <a href="#">View alarms +</a> |

Choose IAM instance profile in the Advanced details.

▼ **Advanced details** [Info](#)

**Domain join directory** | [Info](#)

Select ▼

↻ [Create new directory](#) [↗](#)

**IAM instance profile** | [Info](#)

LabInstanceProfile  
arn:aws:iam::302531438417:instance-profile/LabInstanceProfile ▼

↻ [Create new IAM profile](#) [↗](#)

In the User data, give the UserDataScript-phase-3.sh script.

```
#!/bin/bash -xe
apt update -y
apt install nodejs unzip wget npm mysql-client -y
#wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-200-ACCAP1-1-DEV/code.zip -P /home/ubuntu
wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-200-ACCAP1-1-91571/1-lab-capstone-project-1/code.zip -P /home/ubuntu
cd /home/ubuntu
unzip code.zip -x "resources/codebase_partner/node_modules/*"
cd resources/codebase_partner
npm install aws aws-sdk
export APP_PORT=80
npm start &
```



## Task 6: Migrating the database

- Migrate the data from the original database, which is on an EC2 instance, to the new Amazon RDS database.
- Use *Script-3* from the AWS Cloud9 Scripts file (cloud9-scripts.yml) to migrate the original data into the Amazon RDS database.

```
mysqldump -h 10.0.8.247 -u nodeapp -p --databases STUDENTS > data.sql
```

```
mysql -h database-1.clvflfwpvhti.us-east-1.rds.amazonaws.com -u nodeapp -p STUDENTS < data.sql
```

```
voclabs:~/environment $ aws secretsmanager update-secret \  
> --secret-id Mydbsecret \  
> --secret-string "{\"user\":\"nodeapp\",\"password\":\"student12\",\"host\":\"database-1.clvflfwpvhti.us-east-1.rds.amazonaws.com\",\"db\":\"STUDENTS\"}"  
{  
  "ARN": "arn:aws:secretsmanager:us-east-1:302531438417:secret:Mydbsecret-WnU5WU",  
  "Name": "Mydbsecret",  
  "VersionId": "f6ddfd14-cc74-457b-8bf3-826068e23f50"  
}  
voclabs:~/environment $ mysqldump -h 10.0.8.247 -u nodeapp -p --databases STUDENTS > data.sql  
Enter password:  
voclabs:~/environment $ mysql -h database-1.clvflfwpvhti.us-east-1.rds.amazonaws.com -u nodeapp -p STUDENTS < data.sql  
Enter password:  
voclabs:~/environment $
```



# Phase 4: Implementing high availability and scalability

## • Task 1: Creating an Application Load Balancer

### Basic configuration

**Load balancer name**  
Name must be unique within your AWS account and can't be changed after the load balancer is created.

web-Load balancer

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

**Scheme** | [Info](#)  
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the **IPv4** and **Dualstack** IP address types.

**Load balancer IP address type** | [Info](#)  
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**  
Includes only IPv4 addresses.

# Creating a Load Balancer

- EC2 > Load Balancers > Create Application Load Balancer
- Assign a name to the Load balancer in Basic Configuration.
- In Network Mapping, choose your VPC.
- Under Mappings, choose first availability zone, then choose the Public Subnet.
- Choose the second availability zone, then choose the public Subnet.

The screenshot shows the 'Create Application Load Balancer' wizard in the AWS Management Console. The breadcrumb navigation indicates the path: EC2 > Load balancers > Create Application Load Balancer. The page contains several configuration sections:

- VPC Selection:** A dropdown menu shows 'vpc-02f2bbfad78d58798 (xyz-web-vpc-vpc)' with a '10.0.0.0/16' CIDR block. A 'Create VPC' link is available.
- IP pools:** A section with an 'Info' link. It includes a checkbox 'Use IPAM pool for public IPv4 addresses' which is currently unchecked. A note states: 'The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.'
- Availability Zones and subnets:** A section with an 'Info' link. It instructs to 'Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.'
- Zone 1 Configuration:** The 'us-east-1a (use1-az1)' checkbox is checked. Under 'Subnet', a dropdown menu shows 'subnet-03aa3555d56089d83' with an 'IPv4 subnet CIDR: 10.0.0.0/20'. The selected subnet is 'xyz-web-vpc-subnet-public1-us-east-1a'.
- Zone 2 Configuration:** The 'us-east-1b (use1-az2)' checkbox is checked. Under 'Subnet', a dropdown menu shows 'subnet-00c3f5cf8fcc1f1d9' with an 'IPv4 subnet CIDR: 10.0.16.0/20'. The selected subnet is 'xyz-web-vpc-subnet-public2-us-east-1b'.

# Creating a load Balancer

Create a new Security Group in Security Group

Security group name [Info](#)

LB-SG

Name cannot be edited after creation.

Description [Info](#)

allow http

VPC [Info](#)

vpc-02f2bbfad78d58798 (xyz-web-vpc-vpc)

Inbound rules [Info](#)

Type [Info](#)

HTTP

Protocol [Info](#)

TCP

Port range [Info](#)

80

Source [Info](#)

Anyw...

Description - optional [Info](#)

Delete

# Create Target Group

**Listeners and routing** [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80

Remove

Protocol

HTTP ▼

:

Port

80

1-65535

Default action

Info

Forward to

Select a target group ▼

↺

[Create target group](#)

☰ EC2 > Target groups > Create target group

Protocol

Protocol for load balancer-to-target communication. Can't be modified after creation.

HTTP ▼

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-02f2bbfad78d58798 (xyz-web-vpc-vpc)

10.0.0.0/16

↺

Create VPC

Protocol version

☒ HTTP1

Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

←

↺

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTargetGroup:protocol=HTTP;vpc=vpc-02f2bbfad78d58798

☆

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aws

☰

🔍

Search

[Alt+S]

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❓

⚙

United States (N. Virginia) ▾

Account ID: 3025-3143-8417 ▾

voclabs/user4151272=Rawan\_Hasan

☁

Cloud9

☰

[EC2](#) > [Target groups](#) > Create target group

🏠

🗨

0 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

1 selection is now pending below. Include more or register targets when ready.

Review targets

Targets (1)

🔍

Filter targets

🔴

Show only pending

⏪

1

⏩

⚙

Remove all pending

| Instance ID ▾                         | Name ▾   | Port ▾ | State ▾   | Security groups ▾ | Zone ▾     | Private IPv4 address | Subnet ID ▾              |
|---------------------------------------|----------|--------|-----------|-------------------|------------|----------------------|--------------------------|
| <a href="#">i-0c02ae43c34ff9884</a> 🗑 | Host ec2 | 80     | 🟢 Running | launch web        | us-east-1b | 10.0.28.240          | subnet-00c3f5cf8fcc1f1d9 |

Activate Windows  
Go to Settings to activate Windows.

📺

CloudShell

Feedback

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Cookie preferences



# Creating AMI

EC2 instance > Actions > Image and template > Create image

**Image name:** Host ec2 AMI

EC2

Dashboard

EC2 Global View

Events

Instances

Instances

Instances

Amazon Machine Images (AMIs) (1) Info

Owned by me

Find AMI by attribute or tag

AMI ID = ami-0a773470ae606d35a

Clear filters

Recycle Bin

EC2 Image Builder

Actions

Launch instance from AMI

|                          | Name | AMI name     | AMI ID                | Source                    | Owner        | Visibility |
|--------------------------|------|--------------|-----------------------|---------------------------|--------------|------------|
| <input type="checkbox"/> |      | Host ec2 AMI | ami-0a773470ae606d35a | 302531438417/Host ec2 AMI | 302531438417 | Private    |

## Create a Launch Template

Amazon Machine Image (AMI)

Host ec2 AMI  
ami-0a773470ae606d35a  
2025-10-16T14:53:58.000Z Virtualization: hvm ENA enabled: true Root device type: ebs Boot mode: uefi-preferred

Description

-

Architecture

AMI ID

Storage (volumes)

1 volume(s) - 30 GiB

Free tier:

In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million



# Creating Auto Scaling Group

aws

Search [Alt+S]

United States (N. Virginia)

Account ID: 3025-3143-8417  
voclabs/user4151272=Rawan\_Hasan

Cloud9

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 4 - optional  
Configure group size and scaling

Step 5 - optional  
Add notifications

Step 6 - optional  
Add tags

Step 7  
Review

Enter a name to identify the group.

web-ASG

Must be unique to this account in the current Region and no more than 255 characters.

Launch template [Info](#)

[Switch to launch configuration](#)

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

HOSTec2template

[Create a launch template](#)

Version

Latest (1)

[Create a launch template version](#)

Description

-

Launch template

[HOSTec2template](#)  
lt-0514eec0e5d3a3682

Instance type

t3.micro

AMI ID

ami-0a773470ae606d35a

Security groups

-

Request Spot Instances

No

Key pair name

xyz-web

Security group IDs

[sg-0ea9955cb55c135d5](#)

Activate Windows

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-02f2bbfad78d58798 (xyz-web-vpc-vpc)  
10.0.0.0/16

Create a VPC

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

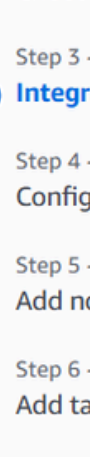
use1-az2 (us-east-1b) | subnet-01570c911ca24e3a1 (xyz-web-vpc-subnet-private2-us-east-1b)  
10.0.144.0/20

use1-az1 (us-east-1a) | subnet-02aea07679fec43ce (xyz-web-vpc-subnet-private1-us-east-1a)  
10.0.128.0/20

Create a subnet

Choose a VPC and for Availability zones and Subnets

Choose *Private Subnet 1* and then choose *Private Subnet 2*. This will launch EC2 instances in private subnets across both Availability Zones.



- Step 2
  - Choose instance launch options
- Step 3 - *optional*
  - Integrate with other services**
- Step 4 - *optional*
  - Configure group size and scaling
- Step 5 - *optional*
  - Add notifications
- Step 6 - *optional*
  - Add tags
- Step 7
  - Review

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

☐ **No load balancer**  
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒ **Attach to an existing load balancer**  
Choose from your existing load balancers.

☐ **Attach to a new load balancer**  
Quickly create a basic load balancer to attach to your Auto Scaling group.

☒ **Choose from your load balancer target groups**  
This option allows you to attach Application, Network, or Gateway Load Balancers.

☐ Choose from Classic Load Balancers

Select target groups ▼ 

Targetgroup | HTTP  
Application Load Balancer: web-Loadbalancer

- Step 5 - optional
- Add notifications
- Step 6 - optional
- Add tags
- Step 7
- Review

Units (number of instances) ▼

Specify your group size.

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity                      Max desired capacity

Max desired capacity

Equal or less than desired capacity

6

Equal or greater than desired capacity

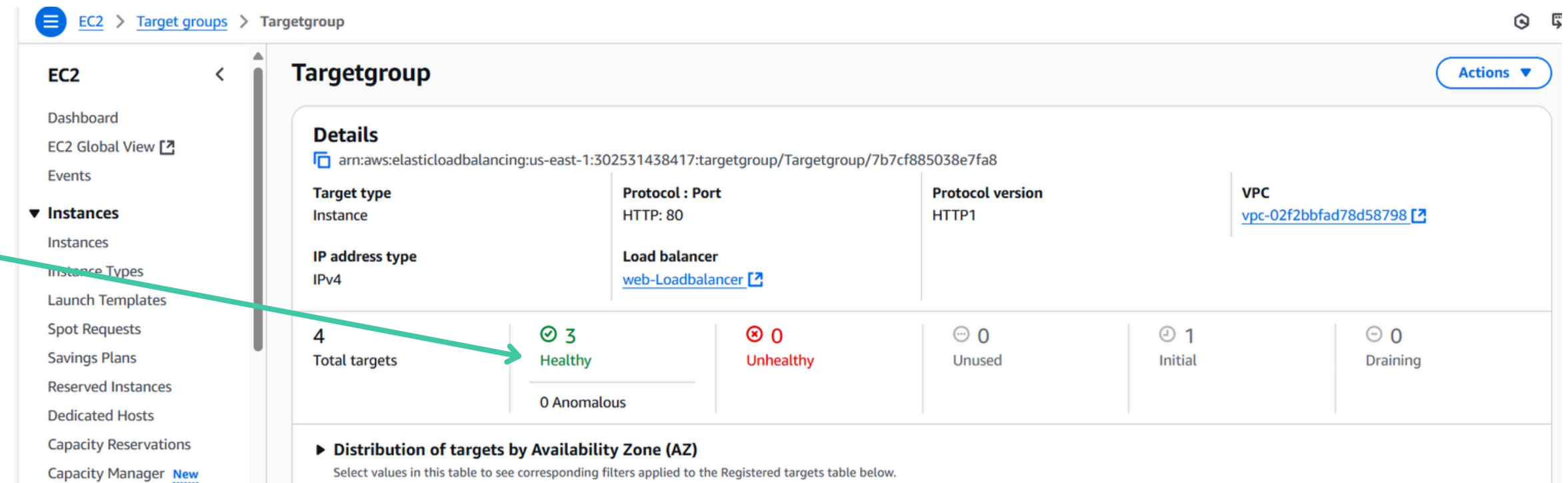
Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☒ No scaling policies

# Testing the Application

The **Health status** column shows the results of the load balancer health check that is performed against the instances.

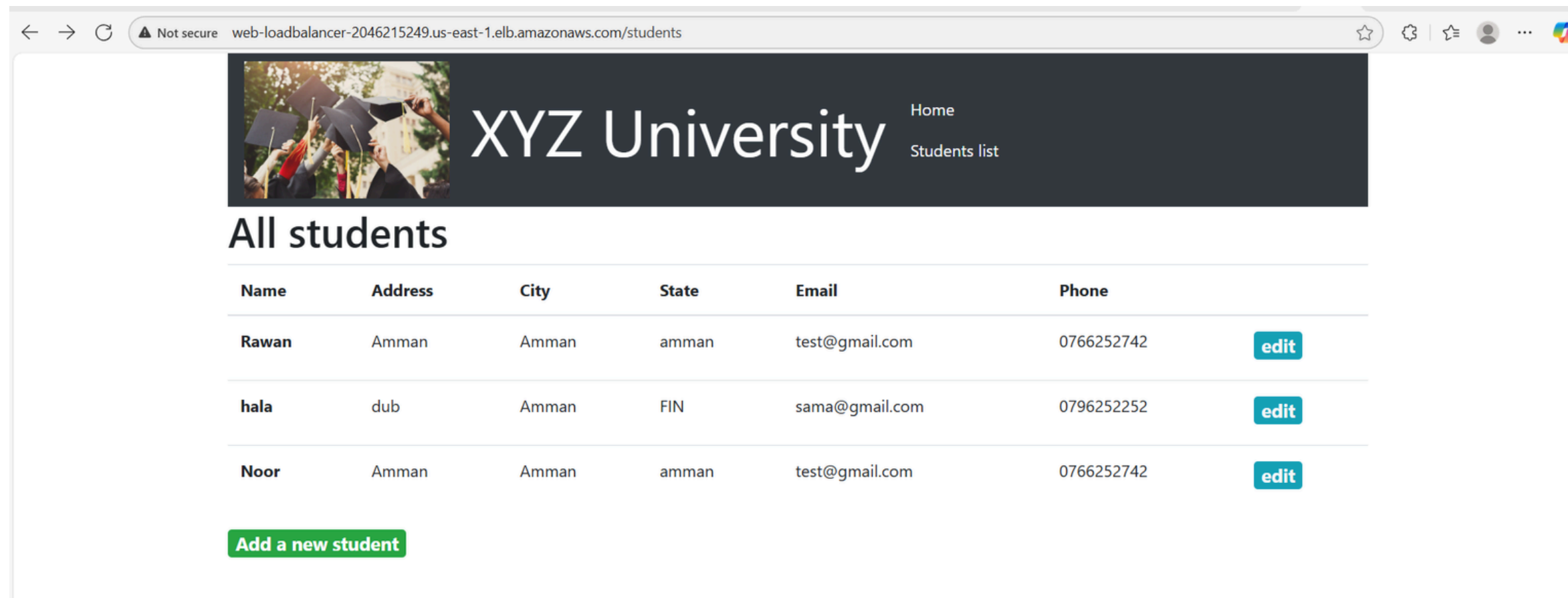


The screenshot displays the AWS Management Console interface for the EC2 Targetgroups section. The left-hand navigation pane lists various EC2-related services, with 'Instances' expanded. The main content area shows the details for a specific Targetgroup. The 'Details' section includes fields for Target type (Instance), IP address type (IPv4), Protocol : Port (HTTP: 80), Protocol version (HTTP1), and VPC (vpc-02f2bbfad78d58798). Below this, a summary row shows the health status of the targets: 4 Total targets, 3 Healthy, 0 Unhealthy, 0 Anomalous, 0 Unused, 1 Initial, and 0 Draining. A green arrow points from the text on the left to the '3 Healthy' status in the summary row.

| Targetgroup                                                                                                                                                         |                                  |                |                  |              |                                       |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------|------------------|--------------|---------------------------------------|--|
| <b>Details</b><br>arn:aws:elasticloadbalancing:us-east-1:302531438417:targetgroup/Targetgroup/7b7cf885038e7fa8                                                      |                                  |                |                  |              |                                       |  |
| Target type                                                                                                                                                         | Protocol : Port                  |                | Protocol version |              | VPC                                   |  |
| Instance                                                                                                                                                            | HTTP: 80                         |                | HTTP1            |              | <a href="#">vpc-02f2bbfad78d58798</a> |  |
| IP address type                                                                                                                                                     | Load balancer                    |                |                  |              |                                       |  |
| IPv4                                                                                                                                                                | <a href="#">web-Loadbalancer</a> |                |                  |              |                                       |  |
| 4<br>Total targets                                                                                                                                                  | 3<br>Healthy                     | 0<br>Unhealthy | 0<br>Unused      | 1<br>Initial | 0<br>Draining                         |  |
| 0 Anomalous                                                                                                                                                         |                                  |                |                  |              |                                       |  |
| <b>Distribution of targets by Availability Zone (AZ)</b><br>Select values in this table to see corresponding filters applied to the Registered targets table below. |                                  |                |                  |              |                                       |  |

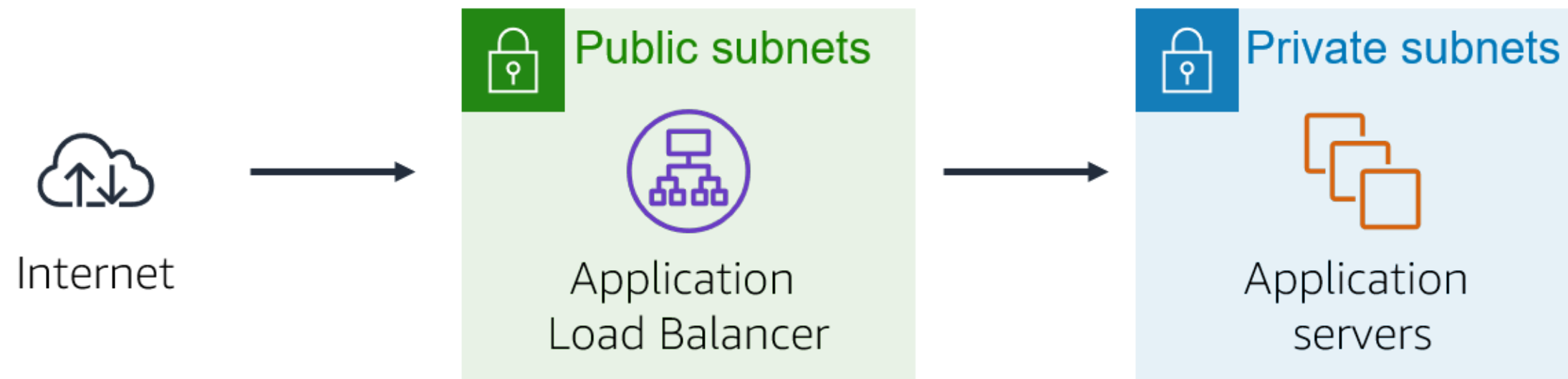
# Testing the Application

In the load balancer details tab copy the DNS name.



You should notice that the instance ID and Availability Zone sometimes toggles between the two instances.



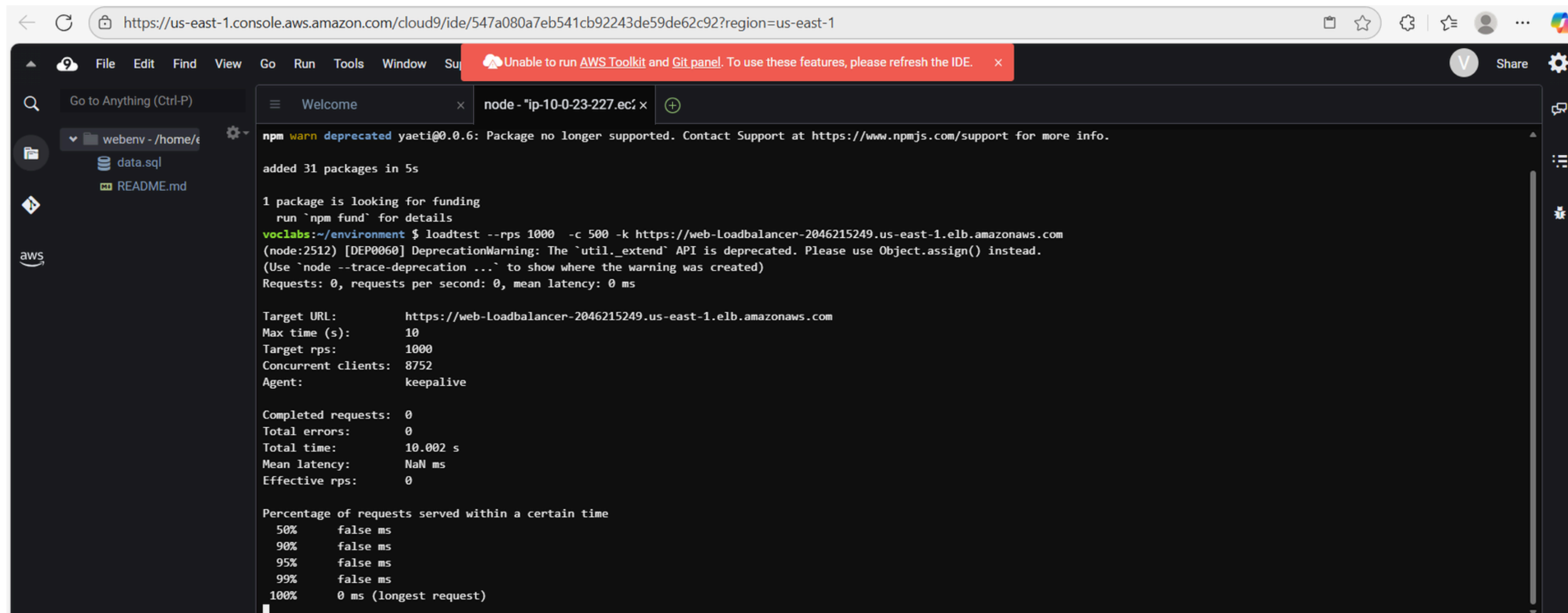


- When this web application displays, the flow of data over the network is shown above.
- You sent the request to the *load balancer*, which resides in the *public subnets* that are connected to the internet.
- The load balancer chose one of the *EC2 instances* that reside in the *private subnets* and forwarded the request to it.
- The EC2 instance then returned the webpage content to the load balancer, which returned it to your web browser.



# Testing the application in Cloud9

On running the Script-2 for load testing using load balancer DNS



The screenshot shows the AWS Cloud9 IDE interface. The browser address bar displays the URL: `https://us-east-1.console.aws.amazon.com/cloud9/ide/547a080a7eb541cb92243de59de62c92?region=us-east-1`. The IDE's top menu bar includes File, Edit, Find, View, Go, Run, Tools, Window, and Search. A red notification banner at the top states: "Unable to run AWS Toolkit and Git panel. To use these features, please refresh the IDE." The left sidebar shows a file explorer with a project named "webenv" containing files "data.sql" and "README.md". The main editor area has a terminal window open with the following output:

```
npm warn deprecated yaeti@0.6: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
added 31 packages in 5s

1 package is looking for funding
  run `npm fund` for details
voclabs:~/environment $ loadtest --rps 1000 -c 500 -k https://web-loadbalancer-2046215249.us-east-1.elb.amazonaws.com
(node:2512) [DEP0060] DeprecationWarning: The `util._extend` API is deprecated. Please use Object.assign() instead.
(Use `node --trace-deprecation ...` to show where the warning was created)
Requests: 0, requests per second: 0, mean latency: 0 ms

Target URL:      https://web-loadbalancer-2046215249.us-east-1.elb.amazonaws.com
Max time (s):    10
Target rps:      1000
Concurrent clients: 8752
Agent:           keepalive

Completed requests: 0
Total errors:     0
Total time:       10.002 s
Mean latency:     NaN ms
Effective rps:    0

Percentage of requests served within a certain time
 50%    false ms
 90%    false ms
 95%    false ms
 99%    false ms
100%    0 ms (longest request)
```

`loadtest --rps 1000 -c 500 -k https://web-loadbalancer-2046215249.us-east-1.elb.amazonaws.com`



**Thank you**